

Shanmukh Yenikapati

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Software Engineer with 5 years of experience building production-scale APIs, agentic AI systems, and full-stack applications. Expert in Python/FastAPI backends, LLM integration (RAG, multi-agent systems, fine-tuning), and rapid prototyping from 0→1. Proven track record of shipping AI-powered products that deliver measurable business impact.

EXPERIENCE

LOWE'S COMPANIES INC

Charlotte, NC

Software Engineer

05/2024 - Present

- Architected and deployed **multi-agent, multi-modal chatbot system** using OpenAI GPT, A2A + MCP protocols, and RAG—boosting containment rate from **50% → 70%** and enabling agentic, context-aware conversations.
- Designed distributed backend services across **10+ microservices** handling **200k+ requests/day**, achieving **30% latency reduction** through Redis caching and optimized routing.
- Built **NLU service** with fine-tuned BERT on FastAPI, improving intent prediction accuracy by **20%** for customer interactions.
- Engineered **Transcript Analyzer** using BERTopic, OpenAI, and Faiss vector indexing for topic similarity search reducing duplicate topic generation by **40%**.
- Owned full service lifecycle from design → deployment → observability (Prometheus + Grafana), and on-call P1/P2 incident resolution; Led rapid prototyping with **LangChain, LlamaIndex, Gemini, and Mistral LLMs** to explore agentic enhancements

Associate Software Engineer

03/2023 - 05/2024

- Engineered data pipelines with **Apache Airflow, PySpark, and GCP Dataproc** to process and transform daily chatbot data from MongoDB into actionable insights, facilitating advanced data analysis and decision-making.
- Created and maintained comprehensive data analytics dashboards in Apache Superset, producing over **200 visualizations** to monitor user behavior, optimize bot flows, and support strategic adjustments.

WALMART GLOBAL TECH

Bentonville, AR

Software Engineer III Intern

06/2022 - 08/2022

- Detected anomalies in DB insert patterns using SQL-driven statistical models, reducing incident tickets by **25%**.
- Automated validation scripts for misconfigured DB properties, cutting manual resolution efforts by **30%**.
- Optimized ingestion pipelines, achieving **18% latency reduction** in batch jobs.

TATA CONSULTANCY SERVICES

Hyderabad, India

Software Engineer

05/2019 - 07/2021

- Built and scaled backend APIs in **Java Spring Boot** processing millions of **healthcare claims daily** with transaction integrity.
- Developed **Spark pipelines in Databricks** powering real-time dashboards for 300+ agents.
- Migrated PL/SQL → T-SQL on Azure SQL for mission-critical workloads, reducing query latency by **40%**.

PROJECTS

Squadfire | Collaborative Task Management Platform | Python, FastAPI, Postgres, HTMX, Tailwind CSS, Docker | [Live](#)

- Built end-to-end task management system: async Python backend, realtime UI with HTMX, gamified leaderboards, and automated task scheduling.
- Designed **task scheduling engine** supporting 4 schedule types (daily/interval/weekly/rotating) with auto-generation via cron jobs.
- Implemented **Task Pools** with bulk import and 4 distribution strategies, enabling users to systematically work through 100+ item backlogs.
- Created **analytics middleware** with GeoIP enrichment, device detection, and GDPR-compliant IP anonymization.
- Deployed on **Oracle Cloud VM with Docker, Nginx reverse proxy, SSL**, and CI/CD pipeline via bash scripts.
- Built **email notification service** with async SMTP, **threaded comments** with @mention support, and dynamic leaderboard image generation (Pillow).

Intelligent Chatbot | Python, Tensorflow, NMT, SQLite, Discord | 12/2020

Developed a Neural Machine Translation-based deep learning model trained on Reddit chat data, integrated a toxicity-filtering score system

IWREN (Improved Wild Relation Network) | Python, Pytorch | 03/2022 | [Repo](#)

Enhanced an abstract visual reasoning model with architecture updates and dataset augmentation, increasing accuracy by 43%

TECHNICAL SKILLS

- Languages & Frameworks:** Python, Java, C/C++, SQL, Go (basic), FastAPI, Flask, Spring Boot
- Data & ML:** Fine-tuning, Model Evaluation, Vector DBs(Faiss), RAG, PySpark, Airflow, TensorFlow, PyTorch, Pandas, OpenAI
- Distributed Systems:** MongoDB, Redis, Kafka, Docker, Kubernetes, Microservices, REST API, GCP (Dataproc, Vertex AI), Prometheus

EDUCATION

RUTGERS UNIVERSITY - NEW BRUNSWICK

New Brunswick, NJ

Master of Science in Computer Science; GPA-3.88/4.0

09/2021 - 05/2023

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

Hyderabad, India

Bachelor of Technology, Mechanical Engineering; GPA-8.32/10

08/2015 - 05/2019

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Design of Internet Services, Software Engineering Web Applications, Database Management Systems, Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), Machine Vision, Probability and Inference, Brain-Inspired Computing